

Ex. No. : 07

Date: 07-04-2026

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A python program to implement decision tree

Aim:

To implement a Decision Tree classifier using Python and analyze classification on the given dataset.

Requirements:

Jupyter Notebook with Python.

PROGRAM:

```
import pandas as pd
import numpy as np
from sklearn.tree import DecisionTreeClassifier
from sklearn import tree
from sklearn.metrics import accuracy_score
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv('gym_statistics.csv')

print("Dataset Preview:")
print(df.head())

# Convert categorical data to numeric
df_encoded = pd.get_dummies(df)
```

```
# Split features and target (last column as target)

X = df_encoded.iloc[:, :-1] y = df_encoded.iloc[:,
-1]

# Train Decision Tree model =

DecisionTreeClassifier(criterion='entropy')

model.fit(X, y)

# Predictions y_pred = model.predict(X)

print("\nPredictions:") print(y_pred)

# Accuracy accuracy = accuracy_score(y,
y_pred) print("\nAccuracy:", accuracy)

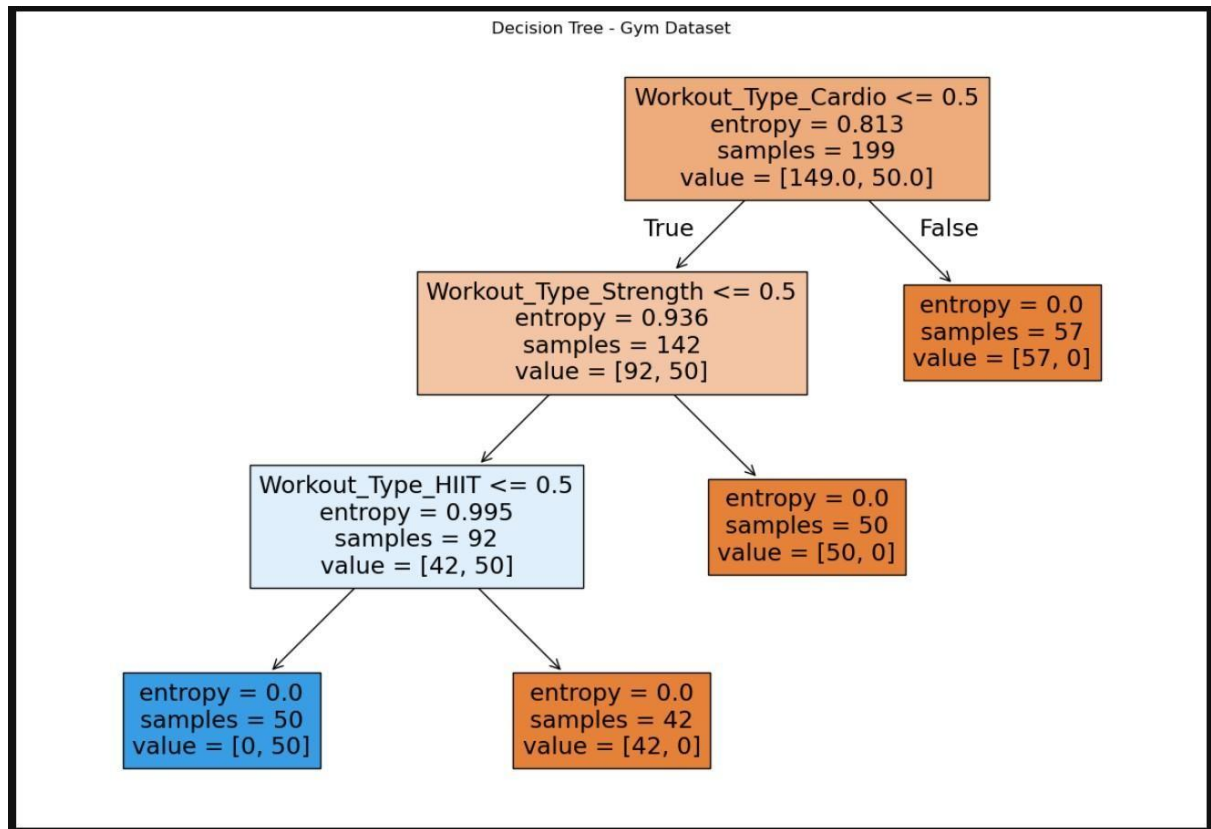
# Plot Decision Tree plt.figure(figsize=(15,10)) tree.plot_tree(model,
feature_names=X.columns, filled=True) plt.title("Decision Tree -
Gym Dataset")

plt.show()
```

Output:

Feature shape: (199, 18)

Target shape: (199,)



Result:

The Decision Tree model was successfully trained and tested, achieving accurate classification on the dataset.